

Introduction to Mathematics

Mathematics is the study of numbers, calculations, shapes, and logical thinking. In Class 4, mathematics helps students understand how numbers are formed, how calculations are done, and how mathematical ideas are used in daily life. Mathematics develops accuracy, problem-solving ability, and clear thinking.

In this class, students learn not only to solve sums, but also to **understand why a method works**. Every calculation follows a step-by-step process, and correct reasoning is as important as the final answer.

Numbers and Place Value

Numbers are made up of digits, and each digit has a value depending on its position. This is called **place value**.

For example, in the number **4,582**:

- 4 is in the thousands place → value = 4,000
- 5 is in the hundreds place → value = 500
- 8 is in the tens place → value = 80
- 2 is in the ones place → value = 2

So,

$$4,582 = 4,000 + 500 + 80 + 2$$

Students learn to:

- read large numbers
- write numbers in expanded form
- compare numbers using greater than ($>$) and less than ($<$)

Example:

Which is greater: **3,456** or **3,465**?

Since $465 > 456$, **3,465 is greater**.

Understanding place value helps students avoid calculation mistakes.

Addition

Addition means **finding the total**. Students learn addition of 3-digit and 4-digit numbers.

Example 1:

$$\begin{array}{r}
 2,345 \\
 + 1,278 \\
 \hline
 3,623
 \end{array}$$

Steps:

- Add ones: $5 + 8 = 13 \rightarrow$ write 3, carry 1
- Add tens: $4 + 7 + 1 = 12 \rightarrow$ write 2, carry 1
- Add hundreds: $3 + 2 + 1 = 6$
- Add thousands: $2 + 1 = 3$

Students also solve **word problems**.

Example:

Ravi has 2,150 marbles. He buys 675 more.

Total marbles = **$2,150 + 675 = 2,825$**

Subtraction

Subtraction means **finding the difference** or **taking away**.

Example 2:

$$\begin{array}{r}
 5,032 \\
 - 1,768 \\
 \hline
 3,264
 \end{array}$$

Borrowing is explained step by step so students understand why it is needed.

Word problem example:

There are 4,500 students in a school.

1,275 students are absent.

Students present = **$4,500 - 1,275 = 3,225$**

Subtraction teaches careful calculation and checking work.

Multiplication

Multiplication is **repeated addition**.

Example:

$$4 \times 6 = 6 + 6 + 6 + 6 = \mathbf{24}$$

Students learn multiplication tables and apply them to larger numbers.

Example 3:

$$\begin{array}{r} 243 \\ \times 4 \\ \hline 972 \end{array}$$

Word problem:

There are 6 boxes. Each box has 18 chocolates.

Total chocolates = **$6 \times 18 = 108$**

Multiplication helps in counting large quantities quickly.

Division

Division means **sharing equally** or **grouping**.

Example:

$$12 \div 3 = 4$$

(12 objects shared among 3 people gives 4 each)

Example 4:

$$84 \div 7 = 12$$

Division with remainder is also taught.

Example:

$$29 \div 5 = 5 \text{ remainder } 4$$

Students learn that:

Dividend = Divisor $\times$ Quotient + Remainder

Division improves logical thinking and fairness concepts.

Fractions

Fractions show **parts of a whole**.

Example:

If a pizza is cut into 4 equal parts and 1 part is taken, the fraction is **$\frac{1}{4}$** .

Students learn:

- numerator (top number)
- denominator (bottom number)

Example:

In **$\frac{3}{5}$** ,

3 = parts taken

5 = total equal parts

Fractions are shown using shapes and objects so students understand visually and numerically.

Time

Time helps students understand daily activities.

Examples:

- 1 hour = 60 minutes
- 2 hours = 120 minutes

If school starts at 9:00 AM and ends at 3:00 PM,
total time = **6 hours**

Students also calculate time differences.

Money

Money problems teach real-life math.

Example:

Cost of a book = ₹135

Cost of a pen = ₹45

Total cost = **₹135 + ₹45 = ₹180**

If you give ₹200,

Change = **₹200 – ₹180 = ₹20**

Money calculations improve practical skills.

Measurement

Students learn:

- Length (cm, m)
- Weight (g, kg)
- Capacity (ml, l)

Example:

2 kg = 2,000 g

Measurement is used in cooking, shopping, and daily tasks.

Geometry and Shapes

Students identify shapes such as squares, rectangles, triangles, and circles.

Example:

A rectangle has:

- 4 sides

- 4 corners
- opposite sides equal

Geometry helps students understand space and structure.

Word Problems and Reasoning

Word problems combine multiple concepts.

Example:

A shop sells 8 packets of biscuits.

Each packet costs ₹25.

Total cost = **$8 \times 25 = ₹200$**

Reasoning questions test understanding, not memorization.

Importance of Mathematics

Mathematics is used in shopping, travel, time planning, and problem solving. Learning maths helps students think clearly, work accurately, and solve real-life problems independently.

By the end of Class 4, students should confidently solve calculations, explain steps, and apply mathematics in daily life.